

DIGITAL LOGIC and DESIGN

Lesson 1 - Introduction to Digital Logic

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Intended Learning Outcome (ILOs):

- After completing this chapter, students will be able to:
- compare the difference between digital and analog.
- understand binary, decimal, octal and hexadecimal system.
- count in binary, decimal, octal and hexadecimal system.
- convert a number from one number system to another system.
- learn how to easily perform binary arithmetic (addition, subtraction, multiplication, and division).

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1.1 ANALOG

Analog solves problems by using continuously changing data such as:

- Temperature
- Pressure
- Voltage

What devices use Analog?

It was the first such device to transmit drawings to a stationary sheet of paper; previous inventions in Europe used rotating drums to make such transmissions. An analog synthesizer is a synthesizer that uses analog circuits and analog **computer** techniques to generate **sound** electronically.

- Old radios
- Megaphones
- Cassette player
- And the volume control on old telephone handsets.

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Advantages of Analog:

- Uses less bandwidth
- More accurate

Disadvantages of Analog:

- High cost of signal conversion inside the display.
- Upgrade to digital interface not possible.

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1.2 DIGITAL

Digital comes from the root *digit*. In Latin, the word “*digitus*” means finger for toe. The modern use of the term *digital* is probably derived from the idea of counting on your fingers. Fundamentally, there are two types of computers such as analog or digital and it describes electronic technology that generates, stores, and processes data in terms of two states: positive and non-positive. Positive is expressed or represented by the number 1 and non-positive by the number 0. Thus, data transmitted or stored with digital technology is expressed as a string of 0's and 1's. Each of these state digits is referred to as a bit (and a string of bits that a computer can address individually as a group is a byte).

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The following are DIGITAL electronic devices include:

- Computers
- Potable media players such as iPods, digital cameras camcorders
- Cellphones
- Radios
- Televisions
- GPSs (Global positioning systems)
- DVD, CD players
- E book readers
- Digital voice recorders
- Handled gaming consoles
- Even cars and appliances, such as microwave ovens, refrigerators, washing machines, include digital electronics for control, monitoring, and fault diagnosis.

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What devices use Digital?

Many people use digital laptops and computers to check their email, browse the internet, and play games. Digital tablets may replace laptops and notebooks in the future. Your phone might be a smartphone that uses digital technology. Modems that are used in computers and smart phones are digital devices.

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1.3 ADVANTAGES AND DISADVANTAGES OF DIGITAL

Advantages of Digital:

- Less expensive
- More reliable
- Easy to manipulate
- Flexible
- Compatibility with other digital system
- Integrated networks

Disadvantages of Digital:

- Sampling Error
- Digital communications require greater bandwidth
- The detection of digital signals requires the communications system to be synchronized

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Table 1. Comparison of Analog and Digital

Factor	Analog	Digital
Signal	Continuous signal which represents physical	Discrete time signals generated by digital modulation
Waves	Denoted by sine waves	Denoted by square waves
Representation	Uses continuous range of value to represent information	Uses discrete or discontinuous values to represent information
Example	Human voice in air, Analog electronic devices	Computers, CDs, DVDs, and other digital electronic devices
Flexibility	Hardware is not flexible	Hardware is flexible in implementation
Uses	Can be used in Analog devices	Best suited for Computing and digital
Memory	Stored in the form of wave signal	Stored in the form of binary bit
Power	Instrument draws large power	Instrument draws only negligible power
Impedance	Low	High order of 100 mega ohm